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SOCIOLOGY | RESEARCH ARTICLE

Making sense of glass ceiling: A bibliometric analysis of conceptual framework, intellectual structure and research publications

Sandeep Singh¹, Chetan Sharma², Purnima Bali³, Shamneesh Sharma² and Mohd Asif Shah^{4*}

Abstract: In life and employment, women face discrimination. The glass ceiling prohibits women from obtaining top management and leadership positions in male-dominated sectors. This study has conducted bibliometric analyses to analyze and highlight the glass ceiling literature. Authors searched international conferences and journals for relevant papers. This study evaluates 1,199 publications from the Scopus database's core collection between 1988 and 2021 to visualize the Glass ceiling. This research analyses co-occurrence, co-citation, and bibliographic coupling to discover the field's most important subjects, authors, and works. The extent of the glass ceiling, current publication patterns, significant regions, key articles, and critical publications were also studied. This research reveals that the glass ceiling is evolving. This research delivers vital insights and conclusions on the glass ceiling's future. Also, rich and developing nations have significant knowledge differences.

Subjects: Art & Gender; Gender and Sexuality in Popular Music; Gender & Sexuality; Women; s & Gender History; Gender and the Law; Religion, Gender and Sexuality; Gender

Keywords: Glass Ceiling; Women; Discrimination; Bibliometric; Scopus; Vosviewer

1. Introduction

Women have been a part of the workforce from time immemorial. In one of the research studies, the researchers observed that almost 50% of the world's total population had been women, which stood at 49.58% in 2019, per United Nations (U.N.) data (D. of E. Affairs and Social, 2019). In recent years, women's participation in the labor market has risen considerably. This influx has forced organizations to adapt their policies, giving women a platform for equality with their male counterparts through legal provisions. These policies aim to provide women with a secure work environment, balancing their work and life and delivering them with facilities in the organizations (The Institute for the Equality of Women and Men, 2013). However, in some of the research studies, it is identified that globally, women's participation in the labor force is 26.5% lower than their male counterparts (ILO, 2018). As a result, authors don't find many women holding top positions in organizations, which further explores why in this century, the struggle is to challenge the existing systems for achieving gender equality (Moghadam et al., 2010). This phenomenon of under-representation of women in higher positions is known as "glass-ceiling, a method summarised in a Wall Street Journal Report on Corporate Women in 1986, 'a barrier so subtle that it is transparent, yet so strong that it prevents women and minorities from moving up in the management hierarchy' (Morrison et al., 1987). The researchers state that Glass Ceiling is a well-known trend in the business world and higher education (Luke, 1997), and women who aspire to higher positions have experienced this (Morrison et al., 1987). The existing studies point to the fact that not only incorporate sector, even women trying to set up their small businesses (Muravyev et al., 2009) or

women who aspire to be entrepreneurs (Bellucci et al., 2010) face bias (Bosse & P. L. T. I. I. I.-T. J. of Applied and Undefined, 2012). It is also observed that being a multidimensional occurrence, the glass ceiling divides organizations and society into gender strata (Ohemeng & Adusah-Karikari, 2014), where the robust structures inhibit the upward movement of women by understating or undermining their talent and intellect (Musoke, 2005). The multidimensional researchers examined that the Glass ceiling as a phenomenon has been studied in various fields like finance (Blau & Kahn, 2007), marketing (De Blas et al., 2012), production (Albrecht et al., 2003), human resources (Pichler et al., 2008), and psychology (Bell et al., 2002) concerning multiple industries. There is a gender disparity in job satisfaction (Singhapakdi et al., 2014), supported by the research that there are also significant gender disparities in lower-order quality of work-life (QWL) and organizational socialization. According to a study, it is concluded that the addition of unique female perspectives and leadership styles will afford gender-diverse TMTs (Top Management Teams) a managerial capability advantage over equally talented yet homogeneous male teams; and this link between gender diversity and organizational capability will be more pronounced in times of crisis (Fernando et al., 2020). The researchers also analyzed women entrepreneurs' motives to survive their businesses (Rey-Marti et al., 2015). It is also implied through the research (Rey-Marti et al., 2015) that women whose motive is to pursue a better work-life balance are less likely to succeed in their professional lives. To step up in their professional lives, women face obstacles identified in the present paper (Briggs et al., 2012) as being excluded from informal networks, a lack of cultural fit, and a lack of mentoring. The mentioned barriers decrease career advancements and put a question mark on the capabilities and skills of working women. A research study, it is examined that the leadership skill of competent women with more work experience is also less likely to aspire to leadership than men. Competent women with less work experience are more likely to aspire to leadership than men (Sanchez & Lehnert, 2019).

2. Literature review

The concept of the glass ceiling has been examined for several decades and has been the subject of a significant number of research papers, reports, and studies, it is impossible to establish the precise date that the first research paper on the glass ceiling was published. However, it is safe to say that the study of the glass ceiling began to acquire significant attention in the late 1980s and early 1990s when more and more research was published on this topic. This is the time period in which there was an increase in the number of studies conducted on this subject. In 1991, the Catalyst organization published a study that was one of the first and one of the most influential ones to be conducted on the glass ceiling (Brady et al., 2011). This study was one of the first to thoroughly investigate women's challenges when trying to advance their careers in corporate America. It also offered a wealth of information and new perspectives on the lives of working women today. Since then, a significant number of research have been carried out to investigate the consequences of the glass ceiling on women and people of color in the workplace. These studies have employed a wide variety of research approaches, which has led to our improved comprehension of the reasons that play a role in the existence of the glass ceiling and the effects that this phenomenon has on individuals and organizations (De Blas et al., 2012; Pichler et al., 2008). As a metaphor, the phrase "glass ceiling" refers to an invisible barrier that stops a specific population (primarily women) from progressing in ranks and being limited to a certain level in an organization. It is also assessed that Feminists initially created the metaphor, as it is popularly used, to describe the obstacles women confront in the workplace. Many nations, including Australia (Maginn, 2010), China (Tan, 2008), France (Barnet-Verzat & Wolff, 2008), South Africa (Booyesen & Nkomo, 2010), the United Kingdom (Thomson et al., 2008), and the United States of America (Eagly et al., 2007), have shown throughout the time that women in leadership roles are underrepresented. The "glass ceiling" metaphor often depicts women's difficulties while advancing their careers and reaching the organization's highest echelons (Burke et al., 2005). It is conceded that there are many theoretical interpretations for the glass ceiling situation (Barreto et al., 2009).

There are too many ideas to examine in-depth on this topic. Still, an effort has been made to utilize the information gleaned from it to put the concepts into practice, creating a foundation on

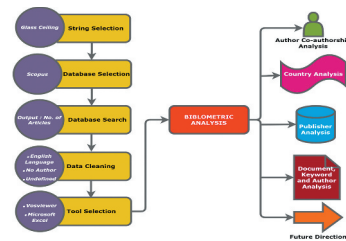
which to work to fill in the holes and inconsistencies. Some research studies also identified that natural selection has resulted in some evolutionary psychologists theorizing the existence of “glass ceilings” as a byproduct of this process (Browne, 2006). Researchers have provided evidence in many studies that there may be a connection between the lack of female leaders and the persistent bias and discrimination women face in the workplace (Weyer, 2007). Examples include women’s exclusion from the old boys club, tokenism, and shadow occupations. Women are exposed to additional scrutiny and a lack of role models that build an unseen package that acts against women, as cited in (Fassinger, 2008). It is common for women who become mothers to face a variety of prejudices that impede their job development and build a maternal wall (Crosby et al., 2004). Gender disparities have been cited as a crucial factor in the rise of gender disparity in leadership. Even though various studies have shown that the “glass ceiling” is a subtle but persistent barrier that is based on discriminatory practices and attitudes, both conscious and unconscious, that prevents qualified women from achieving positions of leadership and management at the highest levels as claimed by one of the researchers (Jackson & O’Callaghan, 2009). Glass ceiling’s manifestation is based on a variety of concepts that have no solid foundation at this time. Because of the pervasiveness of the “glass ceiling,” women are forced to deal with a lack of respect in the job, despite their qualifications, training, and experience.

As per the evidence from various research studies, it can be concluded that women are negatively associated with leadership because they do not aspire to lead in the same manner as their counterparts do. Therefore researchers depict a positive relationship between the proportion of women on TMTs and organizational performance through a research study (Krishnan & Park, 2005). The one more study reveals that gender inequality in a country is relevant. Marital status plays an essential role in defining the position of women on the professional front (Bosch et al., 2018). Researches provided the evidences and elucidated that married U.S. women had a lower likelihood of having a mentor than single women or men (Ramawami et al., 2014). The study highlighted that being married in professional lives disfavoured U.S. women (Bowles et al., 2007). The developments in the social, economic, political, and technology fields in the 21st century have played a role in shaping women’s career choices (O’Neil et al., 2008). Yet, women do not occupy any managerial position in 40% of businesses globally (Thornton, 2021), which has been constant since 2004. On the other hand, women participate actively in economic activities (Adams & Funk, 2012). Still, their roles are restricted due to individual, organizational (Connell, 2006; Dingell & Maloney, 2002), and societal factors (Guy & Schumacher, 2009; Tan, 2008), leading to the income disparity between men and women working in similar positions (Christofides et al., 2013; Eisenberg, 2010). This review is focused on the evolution and spread of the research on the glass ceiling. The objectives of the present study are to identify, integrate and summarize the research work done in the field of glass ceiling through Bibliometric analysis. The study proves its relevance by underlying the importance of the discussions on gender inequality globally and why it is still so important to keep the limelight on this field of research. In addition, the analysis through bibliometrics allows the identification of the focus areas where gaps exist in the current studies. Finally, the study also presents the key indicators of the publications on the glass ceiling. The researchers speak off the phenomenon’s distinctiveness rests in its preponderance at the upper echelons of management (Lupu, 2012). Because of this, it’s accurate to say that women face more significant challenges in the upper echelons than males when it comes to progress. Gendered structures, self, peers, direct managers, family, and society are some of the hurdles that researchers have sought to detect in their study participants (Adams–Harmon & Greer–Williams, 2020).

3. Methodology

An array of tools for analysis is available for quantitative and qualitative review of available literature (Cheng et al., 2018). A scientometric review is a science mapping technique used to analyze (Chen, 2017) and find significant trends and patterns in a specific domain (Cobo et al., 2011). Webometrics/cybernetics studies the web-based presence of information (Thelwall, 2009), like websites, web pages, web content, etc., on a particular topic or field of interest (Björneborn & Ingwersen, 2004). Altmetrics, also called alternative metrics, is the method to gauge the impact of

Figure 1. Proposed Methodology.



a specific topic on social media through analysis of social networking sites (Kwok, 2013) like Twitter and Facebook (Erdt et al., 2016). Finally, bibliometric analysis is a quantitative technique used to search, identify, analyze, and evaluate research publications (Piwowar, 2013).

The visual analysis results, that is, this paper uses CiteSpace and Vosviewer to present the visualization networks, which is helpful to clear the research hotspots and development trends (Chan et al., 2019; Kamdem et al., 2019). The process is essential and has been applied in many areas because of the flexible mapping visualization and powerful user graphic interface (Chen & Liu, 2020; Wang et al., 2020). The methodology used in this study is shown in Figure 1.

Scopus database was used to do Bibliometric analysis. Scopus represents a string of top journals, an internationally recognized benchmark for quality publications. The keywords used to search the database were: “Glass” AND “Ceiling,” “Women” OR “Promotion,” OR “Career” OR “Discrimination.” The results got 1199 publications from 1988 to 2021 (June). However, while searching for relevant papers on the glass ceiling, only one article (Hasan et al., 2021) was slightly related to the current research. So the researchers have done a Scientometric review of the literature. But the paper was only associated with women in the construction industry and contained the analysis from 2000 to 2019. Therefore, the current research focuses on the significance of glass ceiling research by broadly focusing on significant aspects, namely the efficiency and effectiveness of glass ceiling publications based on an array of bibliometric indicators like (i) a total number of publications, (ii) a total number of citations, and (iii) the average citations per publications. Also, using Vosviewer, this paper presents the visual analysis results of the glass ceiling literature by revealing (i) most cited authors, (ii) countries pioneering in glass ceiling research, (iii) institutions focusing on glass ceiling research, (iv) journals carrying glass ceiling research, (v) keywords of research, and (vi) co-authorship (Donthu et al., 2020; Yu et al., 2020).

4. Performance analysis

This systematic review contains analysis based on publication type, research area distribution, years publications, keywords, Country-wise productivity of publications, and other aspects related to the Glass Ceiling. The different analysis aspects are explained below:

4.1. Total publications on glass-ceiling

Figure 2 represents the analysis of publication type and explains that there is a total of 1199 publications were identified from the Scopus database, out of which 920 are articles, 108 are book chapters, 94 are review papers, 45 are conference papers, and only 26 are books, and 06 are short surveys.

Figure 3 represents the analysis based on research areas. The researchers have analyzed that 76.7% of all the publications on the glass ceiling are research articles; the rest all contribute in single-digit percentages. Social Sciences lead in the subject area with 641 publications, followed by 518 in Business, Management, and Accounting and 202 in Economics, Econometrics, and Finance, represented in Figure 3. Since most of the research is multi-disciplinary, total publications are less than when distributed area-wise.

Figure 2. Publication Type.

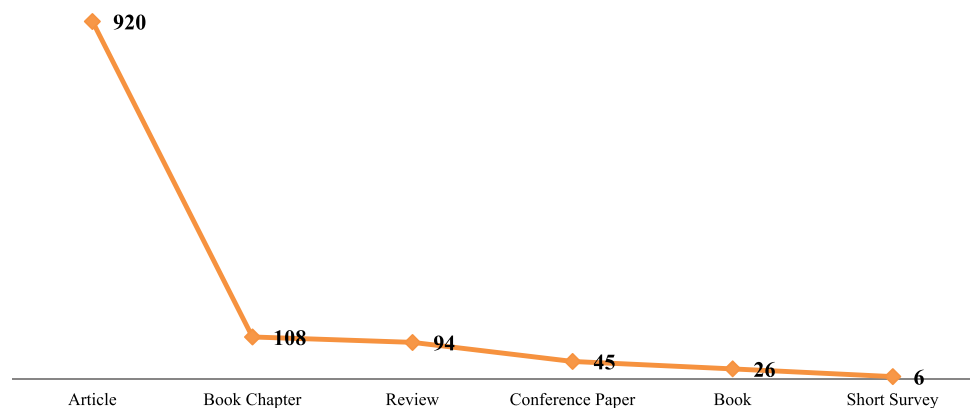


Figure 3. Distribution of Article based on Subject Area.

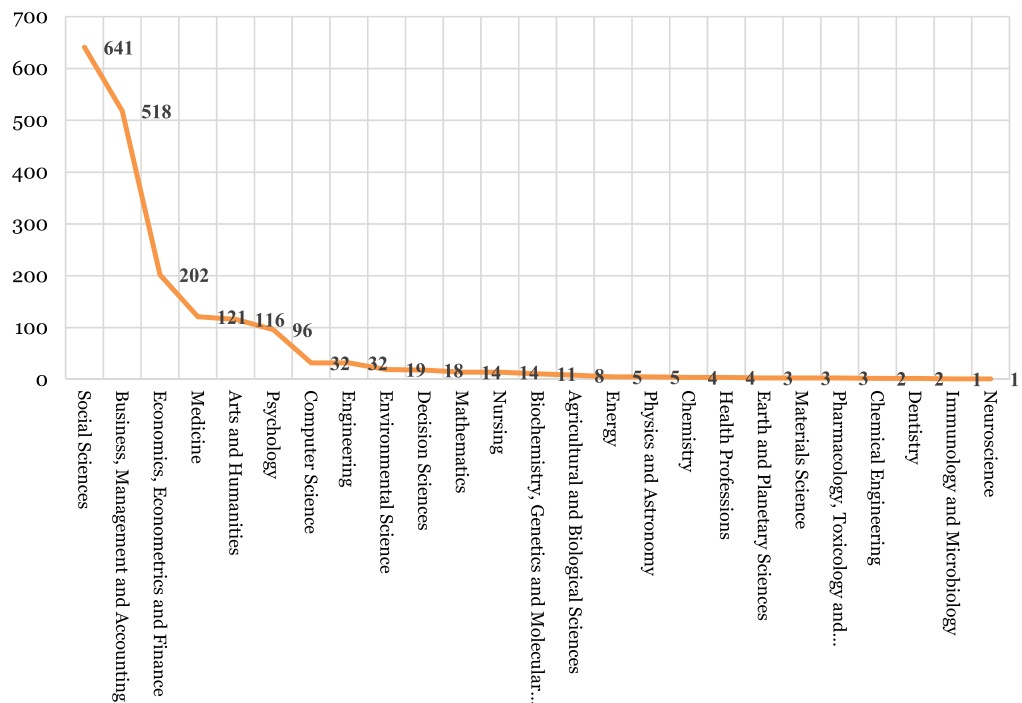
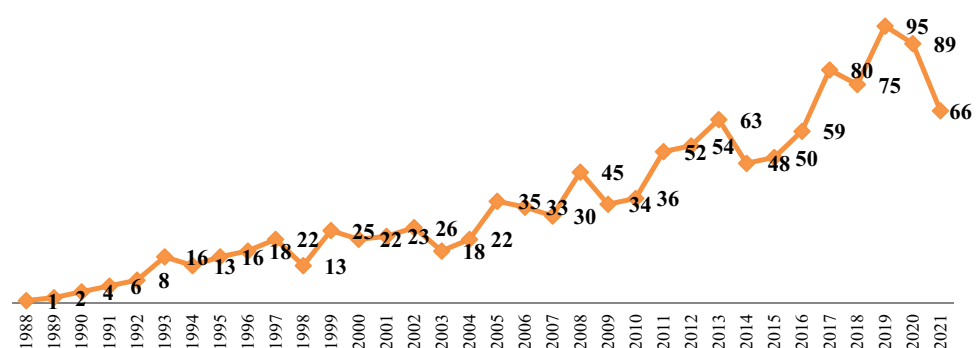


Figure 4. Distribution of Publication based on Year of Publication.



4.2. Year-wise output of Publications

Figure 4 represents a year-wise rising trend since the first paper's publication in the Glass Ceiling field. There may be small dips in the rate of publications between years, like it came down to 13 in 1998, then again to 18 in 2003, but the general trend is towards an overall increase. The year 2019

4.3. Keywords used

Based on the data analysis in Figure 6, the “Women” keyword has 144 occurrences and a link strength of 369. The other keyword, “Leadership,” comes with 123 occurrences and 133 link strengths. There are more than 700 keywords that have been used only once and have zero link strength, like gender stereotypes, gender gap, promotion, women directors, femininity, etc.

Figure 7 represents the network graph of countries based on their research documents. A positive numerical value is assigned for each link, described as link strength. The stronger the connection, the higher this number should be. The amount of times two keywords appear together in a publication is reflected in the total link strength. This analysis states that the United States of America has the most significant publications on the glass ceiling at 427, the highest citations of 9608, and the highest link strength of 38, as shown in Figure 9. A distant second is the United Kingdom, with 91 publications and 2251 citations with a link strength of 24. India is in third place with 65 publications on the glass ceiling, but the citations stand at only 91 and have only four-link power. Australia is in fourth place with 62 publications and 1577 citations, and a link strength of 7. A bibliometric graph may use a distinct hue to represent each country regarding country v/s

A line chart illustrating the decline of the Glass Ceiling index over time. The y-axis represents the index value, ranging from 0 to 300 in increments of 50. The x-axis represents the year, from 1995 to 2020. The index starts at 279 in 1995 and decreases steadily to 34 in 2020. The chart includes labels for various years and categories along the x-axis, such as 'Glass Ceiling', 'Gender', 'Human', 'Female', 'Women', 'Leadership', 'Humans', 'Male', 'Career Mobility', 'United States', 'Glass Ceilings', 'Adult', 'Career', 'Working', 'Discriminator', 'Employment', 'Womens Status', 'Management', 'Physicians, Women', and 'Prejudice'.

Year/Category	Index Value
1995 (Glass Ceiling)	279
2000 (Gender)	214
2005 (Human)	159
2010 (Female)	155
2015 (Women)	144
2020 (Leadership)	123
2025 (Humans)	109
2030 (Male)	91
2035 (Career Mobility)	85
2040 (United States)	62
2045 (Glass Ceilings)	56
2050 (Adult)	54
2055 (Career)	52
2060 (Working)	47
2065 (Discriminator)	42
2070 (Employment)	42
2075 (Womens Status)	42
2080 (Management)	37
2085 (Physicians, Women)	36
2090 (Prejudice)	34

Figure 7. Screenshot of Bibliometric Map Created Based on Countries Document with Network Visualization Mode.



Figure 9. Top 20 Countries based on Citation.

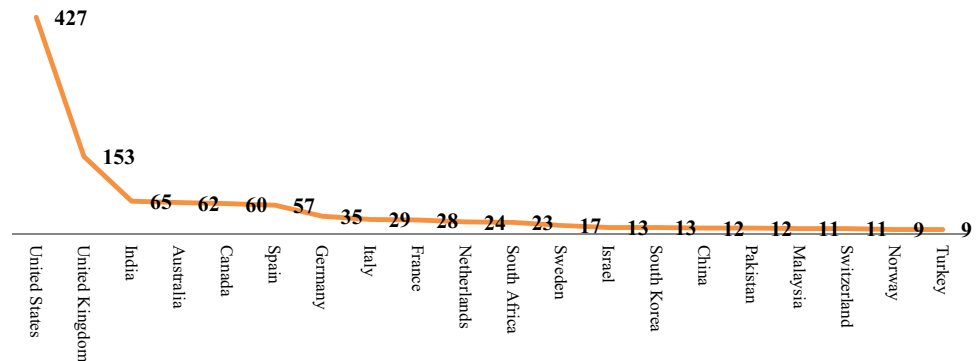
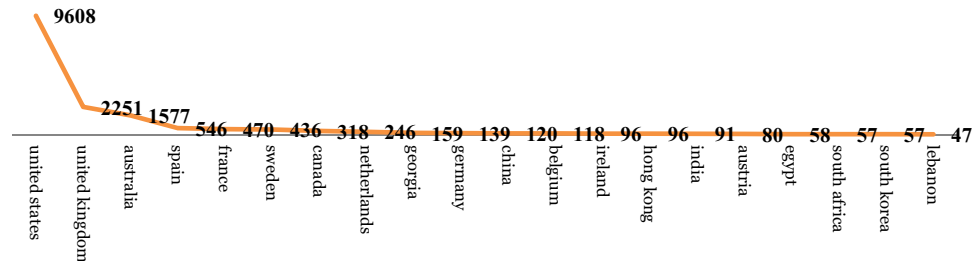


Figure 8. Top 20 Countries based on Publication.



citation analysis. The graph's purpose would be to depict the distribution of research articles published in various countries and how other countries cite them.

Figure 8 supports the analysis of the research Publication made by different countries based on the number of research manuscripts published. According to this comparative structure, India comes at 3rd position in writing papers in the Glass Ceiling field and has a publications strength of 65 from 1988. On the other hand, Canada has 60 publications, 318 citations, and eight link strengths. On the other hand, Spain has 57 publications but stands fourth in citations at 546 and has a link strength of 7.

Figure 9, on the other hand, represents the data of research based on citations. According to this analysis, there are 9606 citations given by different authors to the research manuscripts written by United States researchers. India comes down in this list and has 96 citations preceding the countries like Austria, Egypt, South Africa, and South Korea.

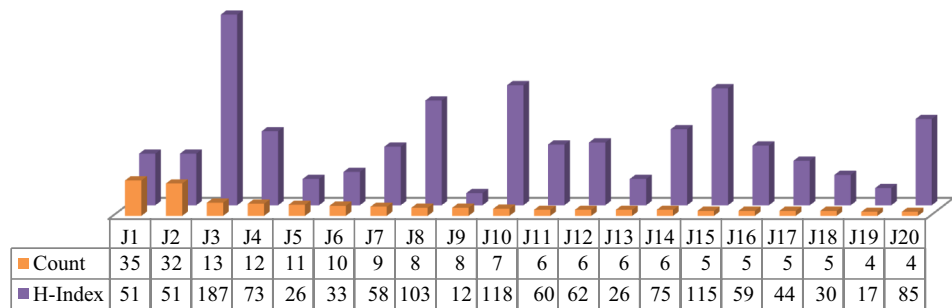
4.5. Journal wise productivity

The network graph for the research publishing based on the Journals is depicted in Figure 10. The link strength of the journals can also be understood through this approach. According to link strengths, the best journals are Gender in Management, Women in Management Review, and Gender and Society, in that order. Gender in Management has 19, Women in Management Review has 12, and Gender and Society has 8. Gender in Management is an international journal that publishes empirical, conceptual, and theoretical articles by researchers and practitioners on gender in management and leadership. The second journal based upon this analysis is "Women in

Figure 10. Screenshot of Bibliometric Map Created Based on Sources with Network Visualization Mode.



Figure 11. Top 20 Journals with their H-Index.



Management Review” which is the previous name of the journal “Gender in Management” so ultimately, combining these two stats, Gender in Management is the journal which is having the highest link strength and “Gender and Society” comes at second place.

The top 20 Journal based on their publication is shown in Figure 11. Women in Management Review (J1) and Gender in Management (J2), both publications of Emerald Group and merged after 2008 as a single journal, are the top journals for publication of research in glass-ceiling with the contribution of 35 and 32 research articles and link strength of 31 and 11 respectively. Journal of Business Ethics (J3) comes at third place with 13 publications and nine link strengths, followed by Gender, Work, and Organization (J4) with 12 publications and seven link strengths; Equality, Diversity, and Inclusion (J5) with 11 publications, and Industrial and Commercial Training with ten publications (J6) (appendix-1).

Figure 12 represents the leading 20 Journals based on the citation. But when it comes to citation, the Journal of Applied Psychology (C1), with only three publications on the glass ceiling, has the highest number of 762 sources, followed by Women in Management Review (C2) at 683 citations; Gender, Work, and Organization (C3) with 624 citations and Journal of Social Issues (C4) with only one publication and 587 citations (appendix-2).

Figure 12. Top 20 Journal based on their Citation.

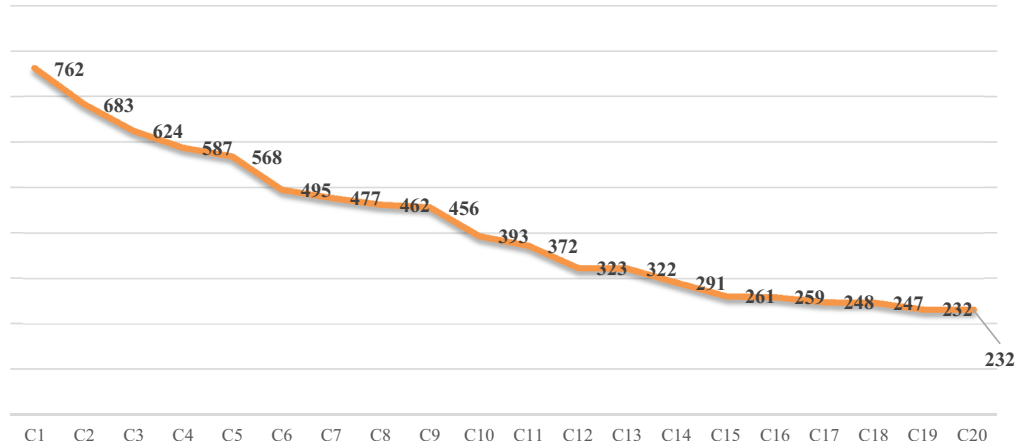
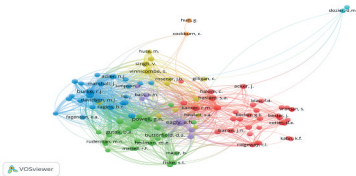


Figure 13. Screenshot of Bibliometric Map Created Based on Most Cited Authors with Network Visualization Mode.



4.6. Author wise productivity

Figure 13 represents the Bibliometric Map based on the most cited authors with Network Visualization. **Burker, R.J.** led the publications on the glass ceiling with seven publications, followed by **Lathabhavan, R.** and **Ryan, M.K.** with six publications, each shown in Figure 14.

Eagle, A.H. has the highest number of 147 citations and a link strength of 2896, followed by **Burker, R.J.** with 124 citations and 3081 link strength. **Powell, G.N.** is in third place with 111 citations and a link strength of 2472, as shown in Figure 15.

Akram, S., Janjua, M., Farugui, N., Martins, R.S., and Zahid, N. are among the top 15 researchers. In addition, they have collaborated with others in glass-ceiling research, all having a total link strength of 15. The network graph of the most-cited authors is shown in Figure 13, and the author’s co-authorship network is shown in Figure 16.

Table 1 lists the most frequently cited articles in Glass Ceiling. The researchers have considered all the papers cited more than 200 times. In addition, the analysis of the H-Index and Impact factor of the Journal where the article was published has also been given in the table. According to this data, **Ridgeway C.L. et al. (2001)** have been published by the *Journal of Social Issues* and cited 587 times by different researchers, whereas **Morrison A.M et al.** has been cited 495 times. The third most cited research paper, “The glass cliff: Evidence that women are over-represented in precarious leadership positions,” has been cited 462 times. Finally, the Journal “Academy of Management Executive” has been renamed to “Academy of Management Perspectives,” which published an article on the topic “Gender gap in the executive suite: CEOs and female executives report on breaking the glass ceiling,” which has been cited 316 times.

Figure 14. Author wise Distribution based on Publication.

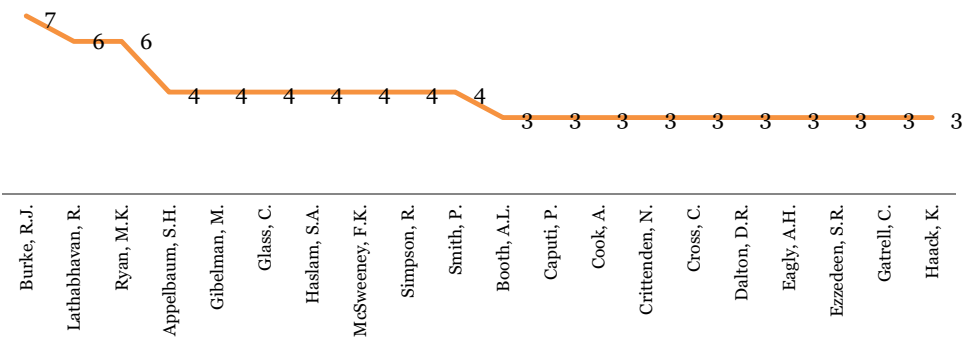


Figure 15. Top Cited Authors.

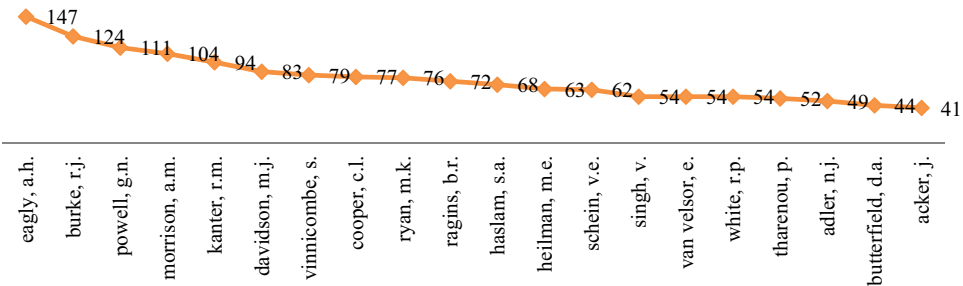
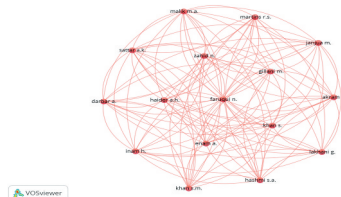


Figure 16. Screenshot of Bibliometric Map Created Based on Author-CoAuthorship Network.



4.7. Institution wise productivity

The York University of Canada has the highest number of publications of 4, followed by Washington State University in the United States at three, and Pennsylvania State University in the United States with three publications shown in Figure 17. The Department of Sociology of Stanford University has the highest citations, 587, followed by the Centre for Creative Leadership, San Diego, with 495 citations, and the University of Southern California, with 495 citations.

Department Of Biostatistics, School Of Public Health, University Of California; Department Of Mathematical Sciences, University Of Texas; Department Of Mathematics And Statistics, Washington Dc.; Department Of Quantitative Health Sciences, University Of Massachusetts; Fay W. Boozman College Of Public Health, the University Of Arkansas For Medical Sciences; Office Of The Provost, Harvard University, Cambridge and Office Of The Provost, North Carolina State University, have the highest link strength of 7 each shown in Figure 18.

5. Intellectual Structure of Glass Ceiling

Table 2 shows the keyword contribution to each cluster and the top keywords for each cluster based on their prevalence in the corpus's different research articles. The fundamental theme of each cluster is determined by the total number of occurrences (TO) of the keywords and the number of other terms to which they are related (DE), as described in Table 3. Based on the analysis presented in Table 2, it can be predicted that there are 11 clusters formed from the corpus based on the significance. Each keyword has its presence in the cluster. The first keyword in each cluster shows the highest contribution to the cluster. For example, the C2K1 has the most significant percentage contribution in the corpus, at 12.1%, whereas the C1K1 has a 10.6% percent advantage. Therefore, the first keyword in each cluster shows the highest contribution to the cluster.

5.1. Cluster 1: women

In the whole corpus, Cluster 1 (women) contributes 25.2% with 35 keywords, out of which 10 have a total occurrence score equal to and more than 5. This cluster has 31.1% connectivity with the keywords of other clusters. "women" is the keyword with the highest importance in this cluster, with 83 occurrences and 219 link strength. Different essential keywords are "leadership", "glass ceiling", "management", "women executives", "equal opportunities", "managers", "women managers," and "barriers". Despite substantial corporate diversity initiatives and regulations promoting equal opportunity for men and women, women remain a minority in decision-making roles. There is an observation that the glass ceiling issue is vertical discrimination against women in organizations (Stamarski & Son Hing, 2015). While the glass ceiling has sparked various studies, some writers claim that theoretical models have done little to grasp the phenomena and its ramifications. The percentage of women in the workforce has steadily risen over the last several years. (e.g., For instance, in Belgium, it was 62.6% at the beginning of 2017, whereas 67% at the beginning of 2019. The influx of women has driven new legal and regulatory measures to secure women's equality with men (Babic & Hansez, 2021).

5.2. Cluster 2: glass ceiling

The second cluster of the corpus contains an 18.7% share. There are 26 keywords in the cluster, out of which ten are mentioned in Tables 2 and 3 based on total occurrence count. "Glass ceiling" is the

Table 1. Top cited research papers in the field of Glass Ceiling

Authors	Title	Year of Publication	Name of the Journal	Total Citations	H-Index of the Journal	Impact Factor
Ridgeway C.L.	Gender, status, and leadership	2001	Journal of Social Issues	587	122	3.424
Morrison A.M.	Women and minorities in management	1990	American Psychologist	495	230	10.885
Ryan M.K.	The glass cliff: Evidence that women are over-represented in precarious leadership positions	2005	British Journal of Management	462	108	6.567
Adams R.B.	Beyond the glass ceiling: Does gender matter?	2012	Management Science	372	255	4.883
Tesch B.J.	Promotion of Women Physicians in Academic Medicine: Glass Ceiling or Sticky Floor?	1995	JAMA: The Journal of the American Medical Association	342	680	56.27
Lyness K.S.	When the fit is fundamental: Performance evaluations and promotions of upper-level female and male managers	2006	Journal of Applied Psychology	325	284	7.429
Reynolds A.	Women in the legislatures and executives of the world: Knocking at the highest glass ceiling	1999	World Politics	323	110	3.025
Ragins B.R.	Gender gap in the executive suite: CEOs and female executives report on breaking the glass ceiling	1998	Academy of Management Executive	316	130	7.846

(Continued)

Table 1. (Continued)

Authors	Title	Year of Publication	Name of the Journal	Total Citations	H-Index of the Journal	Impact Factor
Cotter D.A.	The glass ceiling effect	2001	Social Forces	291	130	1.953
Maume Jr. D.J.	Glass ceilings and glass escalators: Occupational segregation and race and sex differences in managerial promotions	1999	Work and Occupations	281	68	2.276
Daily C.M.	A decade of corporate women: Some progress in the boardroom, none in the executive suite	1999	Strategic Management Journal	259	286	8.641
Lyness K.S.	When fit is fundamental: Performance evaluations and promotions of upper-level female and male managers	1997	Journal of Applied Psychology	256	284	7.429
Probert B.	"I just couldn't fit it in": Gender and unequal outcomes in academic careers	2005	Gender, Work and Organization	249	73	3.101
Zhuge Y.	Is there still a glass ceiling for women in academic surgery?	2011	Annals of Surgery	248	309	12.97
Stuhlmacher A.F.	Gender differences in negotiation outcome: A meta-analysis	1999	Personnel Psychology	232	142	4.362

(Continued)

Table 1. (Continued)						
Authors	Title	Year of Publication	Name of the Journal	Total Citations	H-Index of the Journal	Impact Factor
Korpi W.	Women's opportunities under different family policy constellations: Gender, class, and inequality tradeoffs in western countries re-examined	2013	Social Politics	230	55	1.808
Hakim C.	Women, careers, and work-life preferences	2006	British Journal of Guidance and Counselling	203	46	1.02

Figure 17. Top 20 Institute based on their Publication.

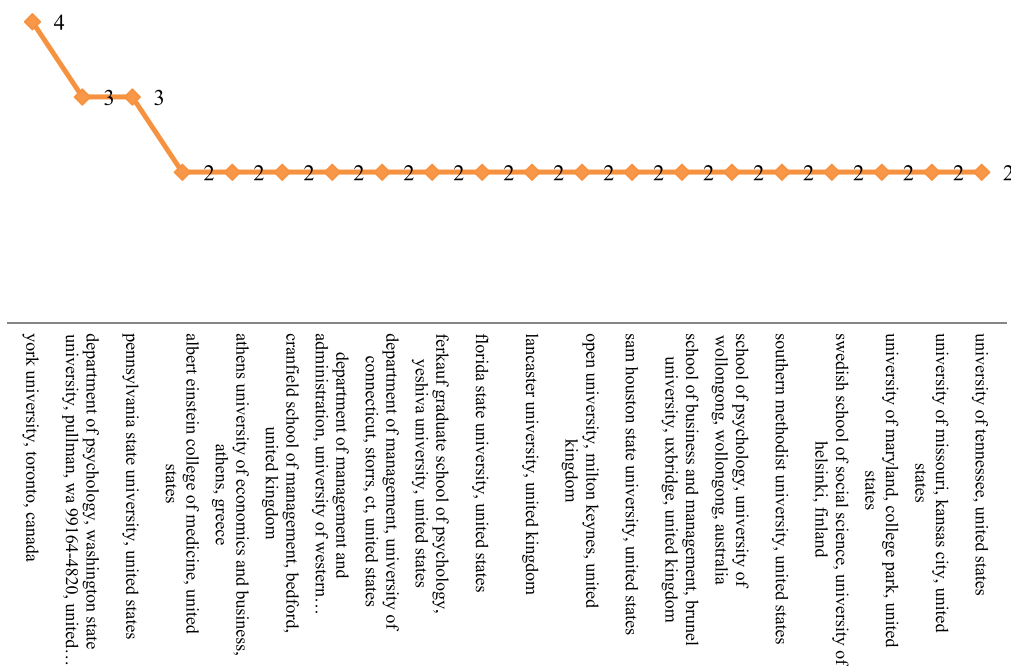
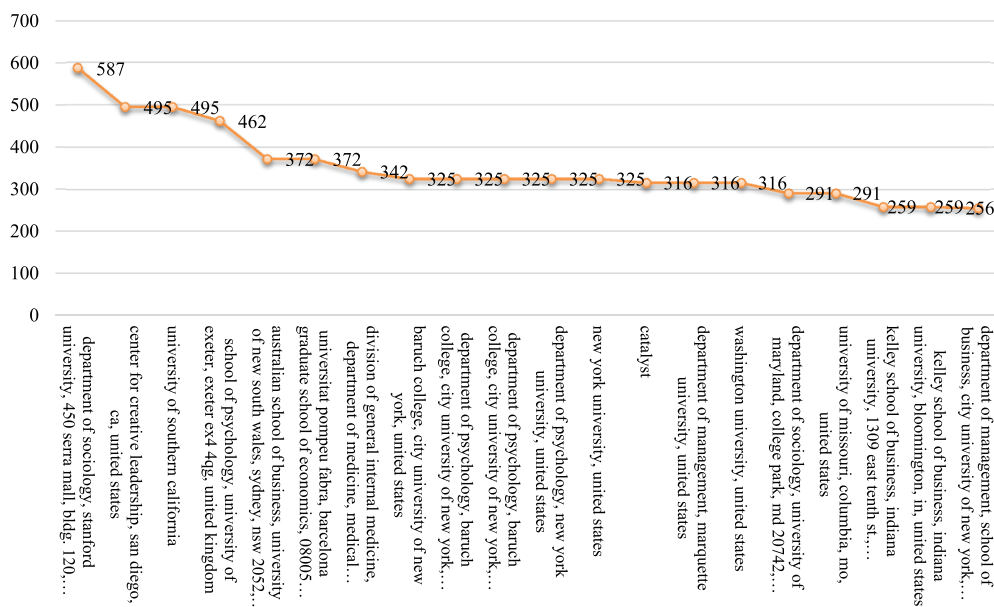


Figure 18. Top 20 Institute based on their Citation.



essential keyword in the cluster, with 95 occurrences and 200 link strength. Other important keywords are “Mentoring”, “Gender Discrimination”, “Gender inequality”, “Higher education”, “Gender bias”, “Career satisfaction”, “Feminism”, “Female leadership,” and “Gender stereotypes”. The “Glass ceiling” in business is commonly accepted as a hindrance for female employees. Invisible hurdles and wooden ceilings keep women out of management. As a consequence, female entrepreneurs confront a second glass ceiling. Escape the glass ceiling by starting up your own business (Tobb et al., 2022). Due to gender prejudices and stereotyping, it is difficult to get financing and establish a new business. Due to patriarchal norms, women are always dependent on their male counterparts (Adikaram & Razik, 2022). It is no longer seen as a matter of social justice or gender equality to focus on women’s

Table 2. Keyword and Cluster Contribution Summary											
Cluster 1			Cluster 2			Cluster 3			Cluster 4		
Keyword	Within Cluster	Within corpus	Keyword	Within Cluster	Within corpus	Keyword	Within Cluster	Within corpus	Keyword	Within Cluster	Within corpus
C1K1	41.5%	10.6%	C2K1	61.7%	12.1%	C3K1	50.6%	9.9%	C4K1	28.1%	2.0%
C1K2	15.5%	4.0%	C2K2	8.4%	1.7%	C3K2	5.8%	1.1%	C4K2	22.8%	1.7%
C1K3	14.0%	3.6%	C2K3	5.8%	1.1%	C3K3	1.9%	0.4%	C4K3	15.8%	1.1%
C1K4	7.5%	1.9%	C2K4	5.2%	1.0%	C3K4	1.9%	0.4%	C4K4	8.8%	0.6%
C1K5	5.5%	1.4%	C2K5	5.2%	1.0%	C3K5	1.9%	0.4%	C4K5	7.0%	0.5%
C1K6	3.5%	0.9%	C2K6	4.5%	0.9%	C3K6	1.9%	0.4%	C4K6	3.5%	0.3%
C1K7	3.5%	0.9%	C2K7	2.6%	0.5%	C3K7	1.9%	0.4%	C4K7	3.5%	0.3%
C1K8	3.5%	0.9%	C2K8	2.6%	0.5%	C3K8	1.9%	0.4%	C4K8	3.5%	0.3%
C1K9	3.0%	0.8%	C2K9	1.9%	0.4%	C3K9	1.9%	0.4%	C4K9	3.5%	0.3%
C1K10	2.5%	0.6%	C2K10	1.9%	0.4%	C3K10	1.9%	0.4%	C4K10	3.5%	0.3%
Performance		25.5%	Performance		19.6%	Performance		14.2%	Performance		7.3%
Cluster 5			Cluster 6			Cluster 7			Cluster 8		
Keyword	Within Cluster	Within corpus	Keyword	Within Cluster	Within corpus	Keyword	Within Cluster	Whole Cluster	Keyword	Within Cluster	Whole Cluster
C5K1	52.6%	2.6%	C6K1	23.8%	0.6%	C7K1	17.6%	0.4%	C8K1	25.0%	0.6%
C5K2	7.9%	0.4%	C6K2	19.0%	0.5%	C7K2	11.8%	0.3%	C8K2	20.0%	0.5%
C5K3	7.9%	0.4%	C6K3	9.5%	0.3%	C7K3	11.8%	0.3%	C8K3	15.0%	0.4%
C5K4	5.3%	0.3%	C6K4	9.5%	0.3%	C7K4	11.8%	0.3%	C8K4	10.0%	0.3%
C5K5	5.3%	0.3%	C6K5	9.5%	0.3%	C7K5	11.8%	0.3%	C8K5	10.0%	0.3%
C5K6	5.3%	0.3%	C6K6	9.5%	0.3%	C7K6	11.8%	0.3%	C8K6	10.0%	0.3%
C5K7	5.3%	0.3%	C6K7	9.5%	0.3%	C7K7	11.8%	0.3%	C8K7	10.0%	0.3%
C5K8	5.3%	0.3%	C6K8	9.5%	0.3%	C7K8	11.8%	0.3%			
C5K9	5.3%	0.3%									
Performance		4.8%	Performance		2.7%	Performance		2.2%	Performance		2.6%

(Continued)

Table2. (Continued)									
Cluster 9			Cluster 10				Cluster 11		
Keyword	Within Cluster	Within corpus	Keyword	Within Cluster	Within corpus	Keyword	Within Cluster	Whole Cluster	
C9K1	26.7%	0.5%	C10K1	35.7%	0.6%	C11K1	46.2%	0.8%	
C9K2	26.7%	0.5%	C10K2	35.7%	0.6%	C11K2	23.1%	0.4%	
C9K3	20.0%	0.4%	C10K3	14.3%	0.3%	C11K3	15.4%	0.3%	
C9K4	13.3%	0.3%	C10K4	14.3%	0.3%	C11K4	15.4%	0.3%	
C9K5	13.3%	0.3%							
Performance		1.9%	Performance		1.8%	Performance		1.7%	

Table 3. Cluster and Keywords based on Highest Occurrence in the Clusters (TO- total occurrences, DE-degree of centrality)

Cluster 1: Women				Cluster 2: Glass Ceiling			Cluster 3: Gender			Cluster 4: Career Development			
Keyword	TO	DE	Keyword	TO	DE	Keyword	TO	DE	Keyword	TO	DE	Keyword	DE
Women	83	219	Glass Ceiling	95	200	Gender	78	178	Career Development	16	50		
Leadership	31	84	Mentoring	13	32	Diversity	9	39	Careers	13	37		
Glass Ceilings	28	82	Gender Discrimination	9	18	Accountants	3	9	Promotion	9	30		
Management	15	39	Gender Inequality	8	14	Boards	3	14	Motivation	5	20		
Women Executives	11	32	Higher Education	8	22	Boards Of Directors	3	13	Gender Gap	4	18		
Equal Opportunities	7	18	Gender Bias	7	12	Brazil	3	8	Career Mobility	2	3		
Managers	7	23	Career Satisfaction	4	18	Ireland	3	8	Egypt	2	7		
Women Managers	7	18	Feminism	4	5	Minorities	3	13	Engineering	2	12		
Barriers	6	17	Female Leadership	3	4	Motherhood	3	7	Information Technology	2	7		
China	5	15	Gender Stereotypes	3	9	Sexual Discrimination	3	11	Russia	2	6		
Cluster 5: Discrimination				Cluster 6: Work			Cluster 7: Work-Life Balance			Cluster 8: United States Of America			
Keyword	TO	DE	Keyword	TO	DE	Keyword	TO	DE	Keyword	TO	DE	Keyword	DE
Discrimination	20	56	Work	5	16	Work-Life Balance	3	11	United States Of America	5	15		
Gender Wage Gap	3	5	Occupational Segregation	4	6	Career Women	2	3	Ethnicity	4	10		
India	3	8	Career Advancement	2	8	Executives	2	4	Entrepreneurs	3	5		
Asian Americans	2	6	Harassment	2	8	Human Resource Management	2	8	Academia	2	4		

(Continued)

Table 3. (Continued)

Hospitality	2	5	Inequality	2	6	Human Resources	2	11	Age	2	7
Race	2	6	Pay	2	10	Organizational Psychology	2	9			
			Intersectionality	2	5						
Sex	2	8	Sexism	2	6	Productivity	2	9	Returns To Education	2	4
Sticky Floor	2	5	Technology	2	1	Wellbeing	2	14			
Wage Discrimination	2	3									
Cluster 9: Australia											
Keyword	TO	DE	Keyword	TO	DE	Keyword	TO	DE	Cluster 11: Gender Equity		
Australia	4	13	Corporate Governance	5	11	Gender Equity	6	12			
Sex Differences	4	10	Gender Diversity	5	11	Participation Of Women	3	7			
Work Engagement	3	13	Board Of Directors	2	4	Applied Behavior Analysis	2	5			
Career Aspirations	2	5	Representative Bureaucracy	2	4	Women Faculty	2	2			
Glass Ceiling Beliefs	2	9									

rights and their contribution to economic progress. For a country to advance economically, it must use its most significant human resource: its women (Shastri et al., 2019).

5.3. Cluster 3: gender

Cluster 3 contains a 15.1% keyword share in the corpus and 17.9% link strength in the keyword co-occurrence network. The keyword “gender” is the most often encountered within this cluster, appearing in 78 articles and linked to another 178 distinct keywords. Other significant words in this cluster are “Diversity”, “Accountants”, “Boards”, “Boards of Directors”, “Brazil”, “Ireland”, “Minorities”, “Motherhood,” and “Sexual discrimination.” Researchers in Brazil and Ireland are working on maternity pay wages. In most economic research, the maternal wage gap compares mothers against non-mothers, defined as women without dependent children (Haynes, 2017). The wage disparity between moms and dads. It is distinct from the gender pay gap, which quantifies the wage disparity between men and women. While the motherhood gap has a sizeable worldwide literature, methodological discrepancies and how mothers, non-mothers, and dads are classified using available data make comparisons challenging. Moreover, in many nations, the data are inappropriate for study due to survey questions that complicate determining a child’s parentage. Nonetheless, several research utilizes worldwide harmonized salary and employment statistics to compare across nations, while others give helpful trend analysis for particular countries (Haynes, 2017). Despite significant gains in studies on women board members and corporate social performance, the emphasis has been on their presence rather than their actual contributions to corporate social responsibility (CSR) activities. Various researchers obtained insights and hypothesized the role of women directors beyond their mere fact on boards of directors and their fundamental contributions via a literature review (Campopiano et al., 2022).

5.4. Cluster 4: career development

This cluster contributes 8.6% of the total keyword volume and has a link strength of 9.5%. “Career development” is the cluster’s most frequent term, occurring 2% of the time in the corpus. This keyword is associated with 50 unique keywords. Other prominent keywords which constitute this cluster are “Careers”, “Promotion”, “Motivation”, “Gender Gap”, “Career Mobility”, “Egypt”, “Engineering”, “Information Technology,” and “Russia”. The keyword “Gender Gap” is an important keyword based on the degree of centrality in this cluster, which moves around all the keywords of the cluster. The gender disparity in participation and accomplishment in science, technology, engineering, and mathematics (STEM) education has been the topic of extensive research for decades (Kayan-Fadlilmula et al., 2022).

According to one of the research studies, 50% of women abandon technology careers by age 35, and women are leaving tech roles at a 45% higher rate than men. Furthermore, only 21% of women in the study said they believed the technology industry was a place they could thrive; sadly, that number falls precipitously to 8% for women of color (Holtzblatt & Marsden, 2022).

5.5. Cluster 5: discrimination

With 6.5% sharing, the cluster “Discrimination” has nine keywords in the corpus. The keyword “Discrimination” had a 52.6% share in the cluster. This keyword has a 2.6% occurrence in about 2.7% of unique keyword linkage in the whole corpus. Therefore, the entire cluster contributes 4.7% in terms of total occurrences. Inequality in pay between men and women is a persistent socio-economic disorder in developed and developing nations. The other two essential keywords in this cluster are “India” and “Gender wage gap,” according to their linkage strength values (Poddar & Mukhopadhyay, 2019). An ADP research found that the gender pay gap has widened during the COVID-19 epidemic, with women missing out on raises and incentives (Times, 2022). In India, just 65% of women earned a wage raise or bonus for taking on more duties or a new function, compared to 70% of men (Nguyen, 2021). Although men and women were equally likely to take on new responsibilities or roles due to COVID-19’s effects on their organizations, the gender gap persists.

5.6. Cluster 6: work

This cluster contributes 5.7% to the whole corpus, with eight keywords. The keyword “Work” has a total contribution of 3.5%, with a total occurrence score of 5. This word is linked with 16 unique keywords in the corpus. The other important keyword in the cluster is “Occupational segregation,” with a 2.8% share in the corpus. Worker distribution is based on demographic parameters, most often gender, and is referred to as “occupational segregation” (Cohen & Huffman, 2003). Few studies have looked at the effect of occupational segregation on advancement opportunities for people of color and women. One research study hypothesizes that initial professional racial and gender composition affects career development in management (Maume, 1999). Men with more female colleagues were likelier to be promoted, whereas Blacks were less likely than Whites to be announced. Women have a lesser chance of becoming managers than blacks. After that, Black men, Black women, and White women had to wait longer for management promotions than White males (Hillman et al., 2002).

5.7. Cluster 7: work-life balance

With an average of 8 keywords, this cluster provides 5.7 % of the whole corpus. The keyword “work-life balance” contributed 2.1% of all occurrences with a score of 3. The corpus associates this keyword with 11 distinct keywords. Women in leadership positions have increased in recent years, but only marginally. Despite growing awareness of the need for workplace equality, multiple studies show that gender-diverse organizations outperform their counterparts. Work-life balance is difficult for many employees, primarily female professionals, and managers. They are often forced to choose between pursuing a potentially promising career, which means giving up maternity leave in the hope of achieving, at most, a position on a board of directors, or starting a family, which means accepting an average job while juggling work and family life (Shankar, 2021). There is a sense that only the active adoption of family-friendly workplace regulations can move the inflexible attitude and push for more equality at all levels (Camilleri & Spiteri, 2021). Some strategies organizations might use to accommodate workers with children better.

5.8. Cluster 8: United States of America

With an average of 5 keywords, this cluster provides 5.7 % of the whole corpus. The Important keywords based upon the occurrence score are “The United States of America,” “ethnicity,” “entrepreneurs,” and “academia”. Regrettably, the academic glass ceiling persists. Only 30% of universities in the United States have female presidents. Diversity in gender in senior management provides several benefits. This study aimed to examine gender roles and the efficacy of leaders in higher education (Dahlvig & Longman, 2014). One new argument for promoting women in academia is that women have often come late. The tension between women’s conventional duties at home and productive work is also said to hinder academic leadership (Chacha, 2021).

5.9. Cluster 9: Australia

This cluster also contributes 5.7 % to the whole corpus with five keywords. The keyword “Australia” contributes 2.8%, with a total occurrence score of 4. This word is linked with 13 unique keywords in the corpus. Another keyword, “Sex difference,” is in the same position as “Australia” in occurrence score but differs in link strength of 10. Six of ten Australian 14–15-year-olds knew what they wanted to do when they grew up (Waters & Flemington, 2021). Most (60%) of those surveyed wanted to work in professional or management positions. There was a strong desire for professional work among those with more significant educational goals, higher accomplishments, or better family socioeconomic standing.

5.10. Cluster 10: corporate governance

In this cluster, there are 4 Keywords, out of which two have equal contributions to the whole corpus. The keyword “Corporate Governance” and “Gender diversity” are directly related to each other and contribute in a way to the main keyword of the corpus, i.e., “Glass Ceiling.” In contrast, the keyword “Board of Directors” and “Representative Bureaucracy” are indirectly related. The board of directors is responsible for setting strategy, high-level structure, and hiring the CEO (Khan

et al., 2022). There is a strong association between board diversity and business innovation, but not between board diversity and board performance (Cumming & Leung, 2021).

5.11. Cluster 11: gender equity

In this cluster, four keywords share 2.8% of the whole corpus. The word keyword “Gender Equity” has a share percentage of 0.8% in the corpus regarding occurrence score. The other keyword which is there in the corpus is “Participation of Women”, “Applied Behavior Analysis,” and “Women Faculty”. Contrary to popular belief, little is known about the role of women in leadership and academia. Westerners write most academic imperialist literature, so it may not truly portray the amount of gender disparity in academia (Jaber, 2022).

6. Limitations of the study

The search results may not show all of the studies available in Scopus connected to the glass ceiling that contain the phrase “Glass” AND “Ceiling”, “Women,” OR “Promotion,” OR “Career” OR “Discrimination”. The reliability of the current study may be compromised, notwithstanding the size of the sample used. Poor retrieval of the literature corpus can be attributed to the limitations imposed by the search words, synonyms, string construction, and variety of search engines used, as well as to the accurate exclusion of results for which the search string is insufficient. Initially, it is important to note that keywords and search strings may rule out relevant functional research. As a next step, authors will use these terms to determine how best to categorize the topics for researchers, scholars, and practitioners. As data is increasing daily, authors may include more data and compare more techniques in the future. The study indicates that a comparison of several databases, such as Scopus and the Web of Science, is planned for the future. One of the examples is “hot paper” feature in the Web of Science search results, which shows the most popular articles. However, there is still something missing from the Scopus feature. It is represented by a fast and significant quantity of quotes, Recognized as soon as published. The more extensive study will be aided by using different data sources that utilize bibliometric analysis. Generally, the more scientific articles are included in future research, the more reliable the general conclusions will be. In addition, identify new challenges and trends in this research area. Therefore, it is imperative to do another quantitative study replicating previously conducted work on the glass ceiling when considering a newly developing area of inquiry. However, the bibliometric analysis may provide clusters as the theoretical basis of glass ceiling activity in a distorted manner due to its inadequacies. Further future research is needed to create an innovative classification system to study trends and developments in the workforce. However, unique bibliometric analysis methods could be used for future investigations since only one VOSviewer application was used in this research.

7. Discussion

Based on the above bibliometric analysis, although discrimination against women has been in the limelight for quite some time, the glass ceiling has not received as much attention from researchers as in other women-related areas. The total publications of only 1199 in 33 years in the Scopus database reflect the general ignorance of social scientists in this field of study; developed countries like USA and U.K share a significant chunk of publications. Most developing countries lag far behind in this field of research. The citations are also high for papers published in developed countries in the same context. Their link strength is stronger than the rest, indicating a robust climate of collaborative research. However, analysis of the keywords suggests a dearth of studies in specific industries or organizations. Most papers give a generalized view of the glass ceiling and do not pinpoint the existence or lack of it in particular sectors/industries/organizations. The maximum average citation per author remains close to 20, again pointing to the low overall productivity of research articles on the glass ceiling. Almost all institutions undertaking glass-ceiling research are from developed countries, with dismissive participation by Asian or African institutions.

8. Implications and conclusion

Based on a study of other literature, it has been concluded that there is a need to strengthen the research on the glass ceiling. Our bibliometric analysis points to gaps that need to be filled so that a comprehensive picture of discriminating gender opportunities can come to the fore. However, this study is based on the keywords generated from the tool's corpus. However, there are still some topics (e.g., women managers, invisible barriers, performance management, discrimination, stereotypes, gender gap, corporate governance, and others) that are still uncovered. This is a threat to the validity of the study, which the authors can cover in the future. As discussed in the result section, the research institutions in developed countries must collaborate with those in underdeveloped and developing countries to understand the need and importance of this research. This paper gives a detailed view of the glass ceiling, and the researchers recommend that in the future, eminent journals should encourage editors to create special issues targeting specific industries, sectors, or countries, thereby providing a publishing platform for good research studies. The funding agencies should also give grants for projects to motivate researchers to pursue the research. Increased participation of women social scientists should also play a defining role in advancing research studies on the glass ceiling. Finally, it needs to find ways to improve the interest of researchers in the field of equal opportunities without any discrimination so that organizations and governments can frame workplace policies according to the needs of the affected people. This paper is an attempt to highlight what needs to be done.

9. Future directions

Comparing the study on the glass ceiling to studies on gender equality, diversity and inclusion in the workplace, and unconscious bias in the workplace will provide a complete view of the progression and development of research on these interrelated topics. Gender equality research addresses the more general issues of gender-based discrimination and equality in the workplace, whereas “glass ceiling” research focuses on the barriers preventing women from attaining top-level positions in the workplace. Diversity and inclusion research studies how organizations may build inclusive workplaces for a wide range of individuals, including women. In contrast, research on unconscious prejudice focuses on the implicit biases that individuals may carry and how these biases can influence workplace decision-making—the research analyses specifically why women are likelier to be promoted than males. When the study of these interrelated topics is compared, it is simple to see how they interact and influence one another. For example, research on the unconscious bias may give light on how implicit biases contribute to the glass ceiling and other barriers to gender equality. Comparing the research completed on various themes can also help identify areas requiring additional study. For instance, if the analysis on implicit bias is more in-depth than the research on the glass ceiling, this may indicate a need for more specific research on the influence of latent biases on women's professional advancement. Comparing the studies on many themes, such as the glass ceiling, gender equality, diversity and inclusion, and unconscious prejudice, can provide a more thorough knowledge of the interrelationships between these problems and guide future study and policy activities.

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